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REVIEW ARTICLE

Remifentanyl for labor analgesia: an evidence-based narrative review

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ABSTRACT

This manuscript reviews the available literature on remifentanyl patient-controlled intravenous analgesia in labor focusing on efficacy and safety. Remifentanyl compares favorably to other potent systemic opioids but with fewer opioid-related neonatal effects. However, remifentanyl provides modest and short-lasting labor analgesia that is consistently inferior when compared to neuraxial analgesia. The initial analgesic effect provided with remifentanyl also diminishes as labor progresses. In several studies, remifentanyl induced significant respiratory depressant effects in laboring women with episodes of desaturation, hypoventilation and even apnea. Given the safety concerns, we recommend that remifentanyl patient-controlled intravenous analgesia should not be a routine analgesia technique during labor. In cases where neuraxial analgesia is refused or contraindicated and the use of remifentanyl justified, continuous and careful monitoring is required to detect respiratory depression to provide safe care of both the pregnant woman and unborn child.

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Introduction

Neuraxial labor analgesia is well established worldwide, especially in developed countries. However, many women, even in the developed world, do not opt for neuraxial analgesia, have no access to this form of pain relief, or have contraindications to its use. Many laboring women therefore receive parenteral opioid-based analgesia using pethidine (meperidine), fentanyl, butorphanol, nalbuphine, tramadol, morphine, alfentanil or sufentanil. Remifentanyl is a potent opioid with pharmacological properties that potentially makes it an ideal parenteral analgesic for labor. For the last two decades, remifentanyl has been studied as either an alternative to neuraxial analgesia or as the preferred parenteral opioid to treat labor pain. Despite the increased use of remifentanyl patient-controlled intravenous analgesia (PCIA) to provide labor analgesia, this indication is considered off-label use. Remifentanyl carries a Category C label (unknown whether the drug can cause fetal harm when administered to a pregnant woman) for use in

pregnancy. Additionally, the manufacturers state “the safety of remifentanyl during labor or delivery has not been demonstrated” and “the drug should be given to a pregnant woman only if clearly needed benefit justifies the potential risk to the fetus.” (Ultiva® (remifentanyl) package insert 2013, Mylan Institutional LLC, Rockford, IL, USA).

For this review, we conducted a literature search with no language restriction on 31 December 2014. Searches were performed in PubMed using the terms “remifentanyl” and “labor analgesia” or “PCIA remifentanyl and labor analgesia”. The results of the search are reported in Table 1. Articles were selected based on relevance to this narrative review on the efficacy and safety of remifentanyl for the provision of labor analgesia. The class of recommendation and level of evidence is stated for each recommendation.

Remifentanyl pharmacokinetics

Remifentanyl has a very short context-sensitive half-time in non-pregnant humans of three minutes,¹ regardless of the duration of infusion. It is rapidly broken down to non-active metabolites by non-specific plasma esterases independent of organ metabolism. Kan et al. demonstrated that remifentanyl rapidly and

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Table 1 Medline search of articles on remifentanyl for labor analgesia

Publication breakdown	Number of papers
Total	135
Not related to topic	33
Basic research	4
Case reports	13
Reviews and meta-analysis	18
Editorials and letters	22
Surveys/audits	6
Trial protocol publication	1
Original clinical studies:	38
Initial feasibility studies	9
Comparison with other opioids	8
Comparison with nitrous oxide	2
Comparison with epidural analgesia	10
Administration modalities	7

extensively crosses the placenta with a uterine vein to maternal artery concentration of 0.88.² In addition, the uterine artery to uterine vein concentration ratio of 0.29 suggests rapid fetal metabolism and/or redistribution. The authors also noted low plasma concentrations in pregnant women following infusion compared to a similar infusion in non-pregnant individuals. This might be explained by increases in remifentanyl clearance, an altered volume of distribution, low plasma protein clearance and an increase in non-specific esterase activity. Jost et al. highlighted large patient variability in pregnant individuals and underlined the need for a pregnancy-specific pharmacokinetic model.³ Ngan Kee et al. confirmed the high trans-placental passage following a bolus at induction of anesthesia for cesarean delivery.⁴

Theoretically, the rapid onset and offset of remifentanyl with effect-site concentration peaking at 1–2 min might be beneficial for labor analgesia, especially if timing of remifentanyl peak effect can be matched to uterine contractions. However, given the duration of a contraction to be 60–80 seconds, the peak remifentanyl effect might only be achieved during the next contraction. Authors have proposed that remifentanyl delivery could be optimized if based on contraction pattern predictions.⁵ Clinically, however, many subjects do not have regular or predictable contractions making timing of remifentanyl peak effect to the contraction peak difficult. The clinically observed large patient variability, underlined in studies like that by Jost, might be partially explained by heterogeneity in uterine contractions as labor progresses.³ The impact of ‘patient-training’ to optimize the timing of remifentanyl administration to coincide with contractions has not been evaluated. Vital-sign integrated drug delivery systems as proposed by Sia et al. may have the potential to improve the safety of remifentanyl PCIA.⁶

Current practice

A wide variety of remifentanyl PCIA administration protocols and settings have been investigated (Table 2). A bolus of 20–40 µg (0.25–0.5 µg/kg) is used most widely with a lockout of 1–5 min. Background infusions are less frequently applied due to maternal side effects and safety concerns, but are occasionally used as they may improve analgesic efficacy.⁷

In Europe, remifentanyl PCIA is increasingly used either as a primary mode of labor analgesia or as an alternative to neuraxial analgesia, when the latter is contraindicated. However, this practice is not uniform over Europe. Saravanakumar et al. surveyed UK practice in 2006 and reported that 50% of units used PCIA opioid analgesia in labor and 35% of these units used remifentanyl PCIA.⁸ Tveit et al. surveyed practice in Norway in 2005 and 2008 and noted that remifentanyl PCIA was used infrequently, in only 2–8% of units, whilst pethidine was used in 77% of units.⁹ One in four women used neuraxial analgesia in this survey. Schnabel et al. surveyed practice in Germany in 2010 and noted that remifentanyl PCIA was only used in 4% of units.¹⁰ A French practice survey by Hanouz et al. in 2011 found that remifentanyl PCIA was used in 52% of units as alternative to neuraxial analgesia when the latter was contraindicated.¹¹ In a Belgian survey of labor analgesia practice, Lavand’homme and Roelants reported that remifentanyl was the first choice opioid alternative when neuraxial analgesia was contraindicated.¹² Since 2009, the use of remifentanyl has grown significantly in Switzerland. Under the patronage of the Swiss Association of Obstetric Anaesthesia, The RemiPCA SAFE Network was established to continuously monitor the use of remifentanyl PCIA in labor nationwide. To date, 24 obstetric units continuously report on all patients in whom remifentanyl PCIA was used either as first-line labor analgesic option or as alternative when neuraxial analgesia was contraindicated.¹³

Analgesic efficacy

Thirty-six original studies, which evaluated the use of remifentanyl PCIA for labor analgesia, were identified from our literature search (Table 1). The breakdown consisted of: feasibility studies ($n=9$); comparisons with other opioids ($n=8$); comparisons with nitrous oxide ($n=2$); comparisons with epidural analgesia ($n=10$); and studies investigating the modalities of administration ($n=7$). Three meta-analysis and systematic reviews were also identified that summarized selected results. The class of recommendation and level of evidence as developed by the ACCF/AHA Task Force on Practice Guidelines are provided.¹⁴

Table 2 Studies of remifentanyl patient-controlled intravenous analgesia in labor

	Bolus	Lockout (min)	Infusion rate ($\mu\text{g/kg/min}$)	Maximum hourly dose (μg)	Number of remifentanyl patients	Alternative analgesia required
Olufolabi 2000 ¹⁹	0.25–0.5 $\mu\text{g/kg}$	2–5	0	NR	4	NR
Blair 2001 ²⁰	0.25–1.0 $\mu\text{g/kg}$	2	0.0–0.05	NR	21	38% (epidural)
Volikas 2001 ³¹	0.5 $\mu\text{g/kg}$	2	0	NR	9	45% (nitrous oxide); 11% (epidural)
Volmanen 2002 ²¹	0.2–0.8 $\mu\text{g/kg}$	1	0	No limit	20	NR
Thurlow 2002 ³²	20 μg	3	0	NR	18	55% (nitrous oxide)
Blair 2005 ³³	40 μg	2	0	NR	20	90% (nitrous oxide)
Volmanen 2005 ²⁶	0.4 $\mu\text{g/kg}$	1	0	No limit	20	NR
Evron 2005 ³⁴	20–70 μg	3	0	No limit	43	11% (epidural)
Volikas 2005 ²²	0.50 $\mu\text{g/kg}$	2	0	NR	50	14% (epidural)
Balki 2007 ⁷	0.25–1 $\mu\text{g/kg}$	2	0.075–0.1	NR	20	5% (epidural)
Volmanen 2008 ⁴⁰	0.1 $\mu\text{g/kg}$	1	0	NR	27	NR
D'Onofrio 2009 ²⁴	0	0	0.025–0.15	NR	205	NR
Douma 2010 ³⁵	40 μg	2	0	1200	52	NR
Douma 2011 ⁴¹	40 μg	2	0	1200	10	10% (epidural)
Volmanen 2011 ⁵¹	0.4 $\mu\text{g/kg}$	1	0	NR	45	NR
Ng 2011 ³⁶	25–30 μg	3.75–4.5	0	500	34	>40% (pethidine & nitrous oxide)
Marwah 2012 ³⁷	0.25 $\mu\text{g/kg}$	2	0.025–0.05	3000 over 4 h	47	6% (epidural)
Ismail 2012 ⁴²	0.1–0.9 $\mu\text{g/kg}$	1	0	NR	380	NR
Tveit 2012 ⁴³	0.15 $\mu\text{g/kg}$	2	0	NR	19	10% (epidural)
Stourac 2012 ⁴⁴	20 (10 μg increases)	3	0	NR	12	NR
Shen 2013 ^{49*}	0.1–0.4 $\mu\text{g/kg}$	2	0.05–0.2	NR	53	11% (epidural)
Stocki 2013 ⁴⁷	20–60 μg	2	0	NR	19	NR
Tveit 2013 ²³	0.15–1.0 $\mu\text{g/kg}$	2	0	0	41	5% (epidural)
Lin 2014 ⁴⁶	0.4 $\mu\text{g/kg}$	5	0.04–0.05	NR	170	NR
Freeman 2015 ⁴⁸	30–40 μg	3	0	NR	402	13% (epidural)

*Comparison of bolus only with continuous infusion. NR: not reported.

Feasibility studies – open label, no comparison groups

The analgesic efficacy of remifentanyl PCIA has been studied for the last 15 years, initially in case reports of parturients with contraindications for neuraxial analgesia.^{15–18} Several early feasibility and dose-finding studies were open, non-blinded studies without placebo or active component comparator. In these studies, remifentanyl PCIA demonstrated analgesic effects, which looked promising in most but not all studies. In 2000, Olufolabi et al. studied four patients in whom manual boluses (0.25–0.5 $\mu\text{g/kg}$) with a lockout of 2–5 min were administered.¹⁹ Pain relief was initially mildly effective in some patients, but pain scores increased in all women as labor progressed. Opioid-related side effects, specifically respiratory sedation, pruritus, sedation, and nausea/vomiting, experienced by all patients, contributed to the investigators' decision to terminate the study prematurely. Blair et al. reported 21 women in whom remifentanyl was made available in escalating doses with or without a background infusion, with a maximum bolus of 1 $\mu\text{g/kg}$ and a maximum infusion of 0.05 $\mu\text{g/kg/min}$.²⁰

A median pain score reduction of 3 (numerical verbal pain score (NVPS) scale of 0–10) was achieved. The authors stated that analgesia was worthwhile but incomplete. Overall, bolus doses of 0.25–0.5 $\mu\text{g/kg}$ were most effective and a background infusion did not significantly improve analgesia. In 2002, Volmanen et al. used remifentanyl PCIA in 20 parturients for a period of 60 min during the first stage of labor. The PCIA bolus dose was increased from 0.2 to 0.8 $\mu\text{g/kg}$ in 0.2 $\mu\text{g/kg}$ increments.²¹ The median effective dose was 0.4 $\mu\text{g/kg}$ and NVPS decreased by a median of 4.2. Volikas et al. studied remifentanyl PCIA using a fixed bolus dose of 0.5 $\mu\text{g/kg}$ with a lockout of 2 min.²² Out of 50 women enrolled in the study, seven converted to epidural analgesia. In the remaining 43 women, pain scores decreased from a visual analog scale (VAS 0–100 mm) of 69 mm to 25 mm (nullipara) and 38 mm (multipara). Tveit et al. enrolled 41 parturients using bolus dose only with a lockout of 2 min.²³ The mean bolus dose was 0.7 $\mu\text{g/kg}$. Mean pain VAS scores decreased from 76 mm to 46 mm; one parturient converted to neuraxial analgesia in this study. D'Onofrio et al. studied 205 parturients in

whom remifentanyl was given as a continuous infusion. The initial infusion of 0.025 µg/kg/min could be increased to a maximum of 0.15 µg/kg/min.²⁴ Mean VAS pain scores decreased from 94 mm to 36 mm at 30 min (no data were presented after that time point). The authors reported minimal maternal and neonatal side effects with 87% of patients satisfied with analgesia.

Recommendation: Remifentanyl provides modest labor analgesia (Recommendation: Class IIB; Level of evidence: B).

Comparison of remifentanyl with nitrous oxide

A recent systematic review of nitrous oxide as labor analgesic concluded that nitrous oxide has modest analgesic effects that are significantly less than epidural analgesia.²⁵ However, studies were of poor quality and many questions remained unanswered. Two studies compared remifentanyl PCIA with nitrous oxide for the provision of labor analgesia. Volmanen et al. randomized 20 parturients to a cross-over study in which remifentanyl PCIA and nitrous oxide inhalation were used each for 20 min during labor with a 20 min wash-out period between the two analgesic interventions.²⁶ Remifentanyl produced a mild reduction in pain scores, whilst nitrous oxide had no effect. This study found that remifentanyl PCIA was superior to nitrous oxide in terms of pain score reduction and most women preferred remifentanyl to nitrous oxide.²⁶ Varposhti et al. compared nitrous oxide to nitrous oxide with a remifentanyl infusion,²⁷ and concluded that the addition of remifentanyl to nitrous oxide produced better analgesia than the use of nitrous oxide alone without an increase in maternal side effects.

Recommendation: Remifentanyl provides more effective analgesia than nitrous oxide (Recommendation: Class IIA; Level of evidence: B).

Comparison of remifentanyl with other parenteral opioids

A variety of opioids have been extensively evaluated for the provision of labor analgesia. These studies consistently show modest analgesia, which is less effective than neuraxial techniques.^{28–30} Remifentanyl has been compared to various other opioid analgesics. Volikas and Male compared pethidine PCIA and remifentanyl PCIA in a small study of 17 parturients.³¹ Pain scores were lower in the remifentanyl group. However, 45% of women receiving remifentanyl also used concurrent nitrous oxide, similar to 62% use of nitrous oxide in the pethidine group. Thurlow et al. studied PCIA remifentanyl and intramuscular pethidine in 36 women.³² Remifentanyl reduced pain scores whilst pethidine did not. However, 56% of women in the remifentanyl group also used nitrous oxide analgesia, and seven women required epidural analgesia later in labor.

In the pethidine group, three women received epidural analgesia. Blair et al. compared remifentanyl PCIA with pethidine PCIA,³³ and noted modest analgesic benefits from remifentanyl. Evron et al. randomized 88 patients to either PCIA remifentanyl or pethidine,³⁴ and concluded that remifentanyl produced better analgesia with fewer side effects. Douma et al. compared remifentanyl PCIA, fentanyl and pethidine in 159 patients and reported mild to moderate analgesia for all three opioids with slight superiority with remifentanyl during the first hour of treatment.³⁵ Ng et al. compared intramuscular pethidine with remifentanyl PCIA and noted maximal pain reduction with remifentanyl at 2 h.³⁶ Remifentanyl produced better pain relief than pethidine. Marwah et al. compared fentanyl with remifentanyl and noted similar, moderate pain relief in both groups.³⁷

Leong et al. performed a systematic review comparing pethidine and remifentanyl PCIA.³⁸ They analyzed seven studies, which included 349 parturients. The main conclusion was that remifentanyl reduced the VAS pain scores more than pethidine. Schnabel et al. came to similar conclusions when performing a meta-analysis comparing remifentanyl, epidural analgesia and analgesia with pethidine. The authors reported better pain VAS, higher maternal satisfaction and less conversion to epidural analgesia with remifentanyl PCIA when compared to pethidine.³⁹

Recommendation: Remifentanyl provides more effective analgesia than pethidine (Recommendation: Class I; Level of evidence: A).

Comparison of remifentanyl with neuraxial analgesia

Several studies have compared remifentanyl PCIA to neuraxial analgesia. Volmanen et al. were the first to compare remifentanyl PCIA to neuraxial analgesia for labor pain relief.⁴⁰ The authors compared epidural levobupivacaine/fentanyl with remifentanyl PCIA, and despite poor pain relief scores reported in the epidural group (lowest reported mean VAS pain scores of 4), epidural analgesia was still superior in terms of analgesia.⁴⁰ Numerous studies have subsequently confirmed the analgesic superiority of epidural analgesia to remifentanyl.^{41–48} Remifentanyl has been compared with labor epidural analgesia in two meta-analyses published in 2012 and 2014.^{39,45} Both concluded that epidural analgesia is the superior labor analgesic option, producing significantly better pain and satisfaction scores. The effect size difference between remifentanyl PCIA and epidural as measured by pain score (0–10 cm) 2 h after initiation is estimated to be 3.0 cm (95% CI 0.7 to 5.2).⁴⁵

Since the publication of these meta-analyses, several papers comparing epidural analgesia with remifentanyl PCIA have been published. Lin et al. performed a retrospective study in 370 patients who received either epidural analgesia or remifentanyl PCIA and noted better quality analgesia with epidural analgesia.⁴⁶ Stocki

et al. recruited 39 women and assigned them to receive either remifentanyl PCIA or epidural analgesia.⁴⁷ Epidural analgesia produced significantly better labor analgesia than remifentanyl, but maternal satisfaction seemed unaffected indicating that many women were satisfied with analgesia despite pain scores remaining relatively high. Freeman et al. performed a multicenter, randomized controlled trial in the Netherlands and again confirmed the analgesic superiority of epidural analgesia compared to remifentanyl.⁴⁸

Recommendation: Neuraxial analgesia provides significantly more effective labor analgesia than remifentanyl PCIA (Recommendation: Class I; Level of evidence: A).

In summary, based on current evidence we can conclude that remifentanyl PCIA provides better analgesic quality than pethidine. However, it is significantly less effective when compared to neuraxial analgesia. Remifentanyl also

provides slightly better analgesia than nitrous oxide. Published data suggest that remifentanyl PCIA used in labor can reduce pain scores from the severe (e.g. 8 out of 10) to the moderate (e.g. 4 out of 10) range (Fig. 1).

It is important to note that in many studies examining remifentanyl PCIA in labor, remifentanyl was not the only analgesic modality utilized. Many patients also used additional methods of pain relief such as inhaled nitrous oxide. Despite initial analgesic efficacy, in many studies pain scores increased as labor progressed. Remifentanyl appears most effective during the first few hours of use, after which pain scores gradually return to baseline pre-analgesia levels.^{39,45} Of note, remifentanyl PCIA was often only studied during a limited part of the labor process (1–3 h) and not continued until delivery. It is likely to be more effective if utilized during the first stage compared to the second stage of labor; and if administered should be initiated early in labor.⁴¹ More intense pain scores, remifentanyl-induced

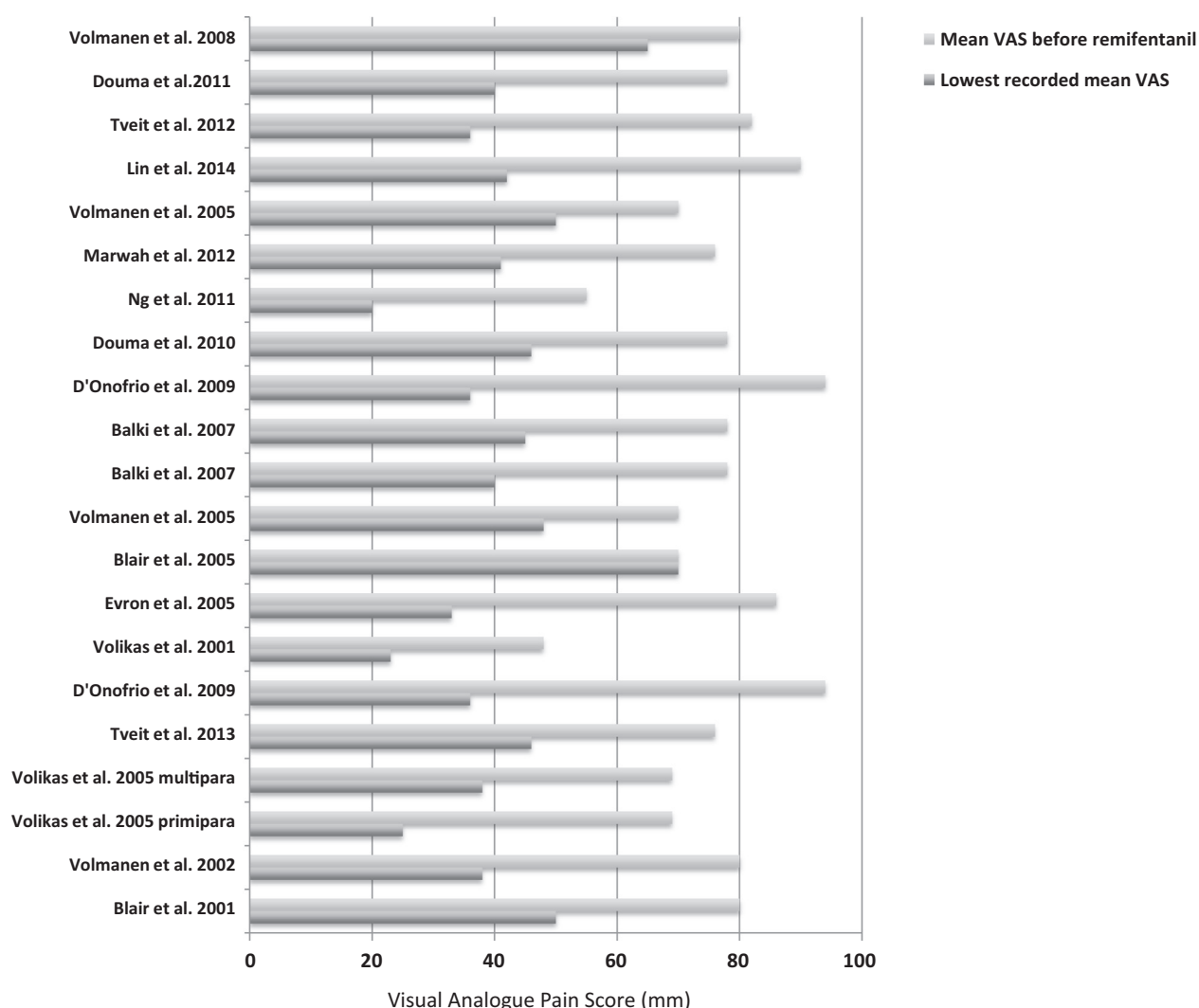


Fig. 1 Pain visual analog score (VAS) at the start of remifentanyl patient-controlled intravenous analgesia (PCIA) and lowest recorded VAS scores during the study period with intravenous remifentanyl PCIA in 21 published studies in laboring parturients. VAS 0–100 mm (0=no pain and 100=worst pain imaginable)

hyperalgesia, or more difficulty timing peak remifentanyl effect with regular and frequent contractions might explain the apparent reduced efficacy in later labor.

Optimizing remifentanyl administration

Protocols vary in the bolus dose, lockout interval, presence or absence of a background infusion, and bolus/infusion administration speed. Published studies show much variability in remifentanyl PCIA settings (Table 2). Of note, the majority of studies do not utilize a background infusion. Balki et al. compared a stepwise increase in background infusion with a constant bolus dose, to a constant background infusion with a stepwise increase in bolus dose.⁷ The authors found similar pain scores but fewer side effects with a stepwise increase in the continuous infusion compared to a stepwise increase in the bolus dose.

On balance, a protocol outlining a fixed bolus of 20–50 µg given on patient demand with a lockout of 1–3 min and no background infusion is a reasonable initial setting. Remifentanyl PCIA settings, however, often need to be adjusted based on analgesic response and labor progress. Lockout intervals can be decreased and bolus size increased as needed. We recommend either not adding a background infusion, or using a small background (0.025–0.05 µg/kg/min) as this may compromise the safety of remifentanyl PCIA.

Shen et al. compared remifentanyl PCIA without background infusion to a continuous infusion of remifentanyl using a blinded and randomized protocol.⁴⁹ In both groups a stepwise increase of the bolus and the infusion was possible. Remifentanyl PCIA provided better labor analgesia than continuous infusion administration. The speed of the patient-controlled bolus is also important. Jost et al.³ compared a ‘modified’ remifentanyl PCIA technique (60 µg bolus infused quickly initially and then tapered-off or stopped if the patient stopped pressing the PCIA button) to a ‘conventional’ group (slower bolus and a bolus dose that started at 20 µg and increased up to 55 µg based on patient analgesic response and side effects). The ‘modified’ method of administration resulted in fewer analgesic requests and lower pain scores as well as fewer changes in PCIA settings compared to the ‘conventional’ group.

Rehberg et al. tried to better predict the peak pain and timing of a future contraction using a handheld dynamometer.⁵⁰ The study aim was to improve the timing of the remifentanyl PCIA bolus to achieve better pain relief with fewer side effects. Although the authors could design a model to predict the peak effect of the next contraction, between-contraction variability was significant, and pain during contractions was not substantially improved when using the models to guide remifentanyl bolus administration. Electrohysterography has also been proposed as a promising parameter to

synchronize labor pain relief with remifentanyl delivery.⁵ Sia et al. are currently developing a system of vital signs-controlled PCIA remifentanyl labor analgesia.⁶ This closed-loop system increases administration of remifentanyl (bolus or/and continuous infusion) based on preset criteria of activation of the PCIA button by the parturient, and decreases remifentanyl if the parturient does not activate the PCIA button. Additionally, the PCIA pump can be stopped automatically if oxygen saturation drops below 95% or heart rate slows to less than 60 beats/min. Volmanen et al. compared immediate delivery of a requested PCIA bolus versus delayed administration in an attempt to better match the remifentanyl peak effect to the maximum effect of a contraction;⁵¹ however, they failed to demonstrate an effect on pain scores.

Safety

Fetal and neonatal safety

Reduced fetal heart rate variability may be observed with remifentanyl infusions. Loss of heart rate variability, a potential indicator of fetal distress, and opioid-induced loss of heart rate variability may create diagnostic confusion in a labor setting.⁵² Data on umbilical artery blood gas values are limited. Similar to other opioids, available data suggest that remifentanyl might induce fetal and neonatal acidosis. Douma et al. reported lower umbilical artery pH values as compared to epidural analgesia, but this did not reach statistical significance.^{35,41} Volikas reported two out of 34 babies with an umbilical artery base deficit greater than 10 mEq/L.²² The need for neonatal resuscitation was lower with remifentanyl compared to other opioids administered in labor (odds ratio 8.6; 95% CI 2 to 34 with remifentanyl compared to fentanyl PCIA in labor).³⁷

Recommendation: Remifentanyl has less effect on the fetus than other parenteral opioids (Recommendation: Class IIb; Level of evidence: B).

Maternal safety

Parenteral opioids can have important side effects for mother, fetus and neonate. With remifentanyl PCIA, numerous studies have found increased maternal sedation, dizziness, nausea, vomiting and pruritus.^{20,22,26,31,33,35,40,51} Douma et al. compared the incidence of maternal fever between parturients without analgesia, epidural analgesia and remifentanyl PCIA. Unlike epidural analgesia, which was associated with maternal fever, remifentanyl was not.⁴¹ Marwah et al. compared remifentanyl with fentanyl for labor analgesia and noted greater oxygen desaturation with remifentanyl (13%) compared to fentanyl (2%).³⁷

The most concerning risk associated with remifentanyl PCIA is maternal respiratory depression. In 2009, Van de Velde noted some degree of respiratory

depression in 32% of laboring patients in whom remifentanyl PCIA was used.⁵³ Several case reports have been published since 2012 in which the administration of intrapartum remifentanyl PCIA resulted in respiratory arrest and/or cardiac arrest.^{54–56} Risk factors for respiratory morbidity were previous opioid administration,^{55,56} or erroneous drug dosing.⁵⁵ However, marked respiratory effects have also occurred without an iatrogenic reason.⁵⁴ Stocki et al. evaluated in detail the respiratory effects of remifentanyl PCIA.⁴⁷ They noted that maternal apnea (defined as >20 seconds of respiratory arrest) occurred in 26% of parturients during the first hour of administration. No apneic events occurred in a comparative group receiving epidural analgesia. They also noted that compared to epidural analgesia, oxygen saturation was lower, end-tidal CO₂ was higher and respiratory and hypoxemia alarms were triggered more frequently. Patients only started breathing again following verbal or physical stimulation. Stocki's study utilizing end-tidal CO₂ and apnea monitors suggests that the incidence of respiratory depression may have been grossly underreported in studies of remifentanyl in labor that only utilized pulse oximetry as the monitor for respiratory depression. Douma et al. noted more sedation and a higher incidence of respiratory depression (defined as a SpO₂ <90%) compared to women receiving epidural analgesia.⁴¹ The incidence of respiratory depression increased from 15% in those receiving epidurals to 48% in women receiving remifentanyl PCIA.

Many authors agree that continuous patient monitoring is required with remifentanyl utilization during labor. Monitoring to determine respiratory depression should ideally consist of continuous one-to-one nursing or midwifery care to evaluate respiratory rate and sedation; monitors to determine adequacy of ventilation (end-tidal CO₂ capnography, apnea monitors); and monitoring maternal oxygenation (pulse oximetry oxygen saturation monitoring) (Table 3).⁵⁷ Supplemental oxygen and methods to manage respiratory depression (e.g. Ambu bag) should be readily available. Routine use of supplemental oxygen is not recommended as it may increase the duration of apnea and reduce the sensitivity of pulse oximetry for detecting hypoventilation. Questions remain whether this degree of monitoring is feasible and economically sound in all patients receiving remifentanyl in labor. An excellent editorial by

Muchatuta and Kinsella analyzed and summarized the risks of respiratory depression.⁵⁸ They highlighted that the risk is real, as evidenced by reported critical incidents associated with remifentanyl use in labor. They also remind us that in the real world setting, continuous monitoring is not always feasible or available. Indeed, they conclude that “nothing we do in anaesthesia is benign; risk is a constant companion to our practice, and must always be considered alongside any perceived benefits. Without due respect for the side-effects of remifentanyl, we risk sleepwalking into a maternal death related to its use in the near future.”⁵⁸

Recommendation: Remifentanyl significantly increases the risk of respiratory depression compared to neuraxial analgesia (Recommendation: Class I; Level of evidence: A)

Conclusion

Remifentanyl compares favorably to other potent systemic opioids but with less opioid-related neonatal effects, and is therefore considered by some the PCIA opioid of choice for the provision of labor analgesia. However, remifentanyl PCIA generally provides modest and short-lasting labor analgesia that is consistently inferior to neuraxial analgesia. The initial analgesic effect provided with remifentanyl PCIA also diminishes as labor progresses. Therefore, remifentanyl PCIA should not be viewed as a comparable alternative to neuraxial analgesia for the provision of labor pain relief. In several studies remifentanyl PCIA induced significant respiratory depressant effects in laboring women with episodes of desaturation, hypoventilation and even apnea. Several case reports of respiratory arrest have also been reported. Given the safety concerns with remifentanyl utilized in labor and clear evidence that neuraxial techniques provide superior labor analgesia compared to any opioid PCIA technique, remifentanyl PCIA should not be considered a routine analgesia technique during labor. Remifentanyl should only be used in isolated cases where neuraxial analgesia is contraindicated or where strong maternal preference to avoid neuraxial analgesic techniques exists. We suggest that the widespread use of PCIA remifentanyl cannot be recommended before safety is established following properly powered and well-designed studies have been performed. In cases where remifentanyl PCIA is used, continuous and careful monitoring is required to detect respiratory depression and to provide safe care of the pregnant woman and unborn child.

Disclosure

Brendan Carvalho has received funding for research from Covidien LP and is a Smiths Medical pain advisory panel member. Marc Van de Velde has previously

Table 3 Mandatory monitoring during remifentanyl PCIA labor analgesia

Continuous uninterrupted one-to-one midwifery/nursing care and surveillance
Monitoring for adequate ventilation: apnea monitoring or capnography
Continuous maternal oxygen saturation monitoring

received funding for lectures from Abbvie, Airliquide, AstraZeneca, Baxter, BBraun, Fresenius, MSD, Smiths Medical and is a Smiths Medical pain advisory panel member.

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